

PRIMA AJI SETYAWAN

Semarang, Indonesia

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Professional Summary

Electronic Engineering Technology student specializing in Embedded Systems, Industrial Automation, Internet of Things, and Control Systems. Experienced in developing microcontroller solutions using ESP32, Arduino, PLC, C/C++, and Python. Proven leadership in robotics research and development through POLIREVO and recognized internationally with a Silver Medal at Indonesia International Invention Expo 2025. Seeking internship opportunities in Automation, Embedded Systems, IoT, and Instrumentation Engineering.

Technical Skills

Programming: C/C++, Python, JavaScript, PLC Programming

Embedded Systems: ESP32, ESP32-S3, ESP8266, Arduino Uno/Nano, Raspberry Pi

Automation & Control: PID Control, Industrial Automation, Instrumentation, Mitsubishi Servo Systems, Control Engineering

IoT & Communication: MQTT, REST API, Firebase Realtime Database, Blynk IoT, WhatsApp API

Engineering Tools: MATLAB Simulink, VS Code, Cursor, GitHub Copilot, Arduino IDE, EasyEDA, Git, GitHub

Project Experience

Post-Stroke Smart Hand Gloves 2024–2025

- Developed an IoT-enabled rehabilitation glove using ESP32-S3 with cloud-based monitoring.
- Designed an LVGL touchscreen interface and integrated database storage with WhatsApp API notifications.
- Awarded Silver Medal at Indonesia International Invention Expo 2025 and 3rd Place at Polines Elektro Expo 2024.

AC Servo Trainer Kit Reconditioning 2026

- Reconditioned and configured an industrial AC servo trainer kit for laboratory applications.
- Performed wiring, PLC programming, parameter setup, troubleshooting, and performance validation on Mitsubishi MR-J2S-10A.

Transporter Robot 2024

- Developed a holonomic transporter robot controlled via Bluetooth using a PS3 controller.
- Programmed an ESP32 to map analog stick inputs to omni-directional kinematic equations for payload delivery.

Water Level Control System Using PID 2023

- Developed a PID-based water level control system using Arduino Uno and MATLAB Simulink.
- Integrated VL6180X sensors for real-time feedback to achieve stable control performance.

Automated Poultry Farm Monitoring System 2023

- Developed an IoT monitoring and PID control system for temperature and humidity regulation using ESP32.
- Implemented MQTT and Node-RED for real-time environmental monitoring and automation.

Leadership Experience

POLIREVO Polines Semarang, Indonesia

Head of KRTMI Division & Programmer Aug 2024–Present

- Led the KRTMI robotics division and programmed autonomous and remote-controlled systems for national competitions.
- Coordinated multidisciplinary teams in software, electronics, and mechanical integration.
- Developed autonomous navigation algorithms and control systems while conducting testing and parameter optimization.

Education

Politeknik Negeri Semarang
D4 Teknologi Rekayasa Elektronika (Semester 6)
Sep 2023–Expected Sep 2027
GPA: 3.47/4.00
Relevant Coursework: Digital Control Systems, Microcontrollers, Programming, Electrical Circuits, Mechatronics, Industrial Automation

Semarang, Indonesia

Awards & Certifications

- Silver Medal, Indonesia International Invention Expo, IoT & Health Technology Category (2025)
- 3rd Place, Polines Elektro Expo, IoT Innovation Category (2024)
- Google Cloud Skills Boost: AI Study Jam Season 12 (2026)
- Belajar Dasar AI Dicoding